Reducing Cross-Node Network Calls in Distributed Payment Systems

B.Tech Project — End-Term Evaluation

with Concurrency-Safe Resolution of Single-Resource Contention

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Outline

The Problem We Started With

Distributed payments at scale

- $\sim$68,000 cross-node operations per second
- Each request triggers **5–15 cross-node calls**
- p95 = 98 ms, p99 = 285 ms at peak
- Strict correctness: every transaction must be exactly once

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Tension

Aggressive caching reduces latency **but** risks staleness, lost updates, and double-allocation in a financial system.

What the Minor Established

- Two-tier in-process caching (thread + server-local) is **safe** for configuration-class data
- Formal cache-safety boundary framework (4 criteria)
- Phase 1 results in production:
  - **4.1%** cross-node call reduction at peak
  - **−11.2%** p95, **−13.0%** p99 API latency
  - **−26.2%** Redis p99 latency
  - Zero correctness incidents

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Open question raised at the minor evaluation

"If you cache state, how do you prevent two users from booking the same single piece?"

This is the Question We Answer

End-term contribution

A **contention layer** that composes with the cache and provably prevents double-allocation on single, indivisible resources — inventory units, reservation slots, account balances, gateway tokens.

Five complementary primitives:

1. Distributed locking with fencing tokens
2. Atomic decrement via Redis Lua scripts
3. Optimistic concurrency control (versioned cache)
4. PN-Counter CRDTs (eventually consistent)
5. Single-flight request coalescing

Two-Tier Cache — Quick Recap

Tier 1: Thread-level

- `IORef HashMap`, per-request
- Destroyed on request end
- **Zero staleness** guarantee
- Hit rate: 22.3%

Tier 2: Server-local

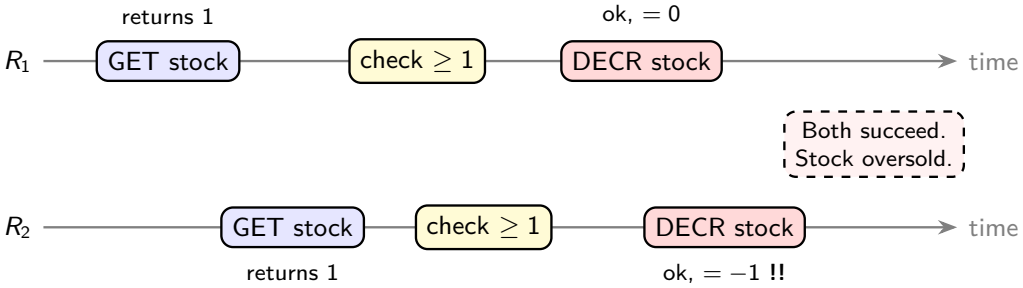
- `TVar` + LRU + TTL
- Shared across threads on a node
- TTLs: 30–120 s by data class
- Hit rate: 41.7%

Cache-safety boundaries (must hold for cache-eligibility)

1. Immutability within TTL
2. Request-scope independence
3. Idempotency of stale reads
4. Deterministic key

Why Caching Alone Is Not Enough

Single-resource state fails criterion (3): a stale read can lead to double-allocation.



The contention layer must close every variant of this race.

Primitive 1 — Distributed Locking with Fencing Tokens

- Acquire a per-resource lock (Redlock on Redis, etcd lease) before read-modify-write
- **Fence token:** monotonically increasing number returned with the lock; backend rejects stale-token writes
- Mitigates GC pauses, partitions, clock drift

Sketch

```
withLock conn key ttl $ \tok -> do
  v   <- backendRead key
  v'  <- compute v
  backendWriteIfFence key v' tok    -- backend checks tok
```

Use when: business logic is non-trivial; contention is moderate; the critical section calls external services.

Primitive 2 — Atomic Decrement via Redis Lua

Idea: package the entire "check + decrement" in one Lua script. Redis runs scripts atomically.

Lua

```lua
local a = tonumber(redis.call('GET', KEYS[1]) or '0')
local w = tonumber(ARGV[1])
if a >= w then
  redis.call('DECRBY', KEYS[1], w)
  return {1, a - w}
else
  return {0, a}
end
```

- Latency: <1 ms; throughput: $> 10^5$ scripts/s/shard
- Cache: **display reads** stay cached; **authoritative decrement** bypasses cache and publishes invalidation
- Limitation: critical section must fit in a Lua script (no external calls)

Primitive 3 — OCC with Versioned Cache Entries

- Each cache entry carries a version
- Reader records the version; writer issues a CAS (`WHERE version = ?`)
- Conflict $\Rightarrow$ retry

SQL form

```
UPDATE inventory
   SET stock = stock - 1, version = version + 1
 WHERE sku = $1 AND version = $2 AND stock >= 1
RETURNING stock, version;
```

Use when: contention is rare (e.g. control-plane writes).
Avoid when: contention is high — retry storms collapse throughput.

Primitive 4 — PN-Counter CRDTs

- Two grow-only counters per replica: P_i (positive), N_i (negative)
- Value $= \sum_i P_i - \sum_i N_i$; merge by element-wise max
- Writes are **local**, never blocked, eventually converge

When to use

Soft holds, cart abandonment counters, flash sales advertising "$\sim$1,000 units" — where a small overshoot is business-acceptable.

When not to use

Hard limits where every unit must be exact. CRDTs trade hard upper bounds for availability.

Primitive 5 — Single-Flight (Request Coalescing)

Idea: on a cache miss, only the **first** request fetches; concurrent requests for the same key wait on the leader's promise.

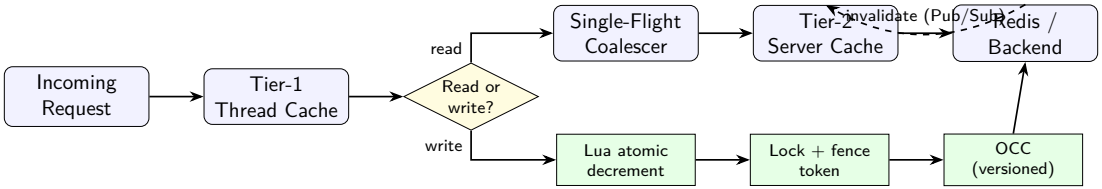
Effect

- Collapses the *front* of the contention curve
- One backend call instead of k identical ones
- Tames cache stampedes / thundering herds

Important

Single-flight **deduplicates** on a single node; it is **not** a substitute for atomicity. Compose it with Primitive 2 (Lua) for full safety across nodes.

Putting It Together



Reads go through cache + single-flight.
Authoritative writes bypass the cache and select a primitive per data class.

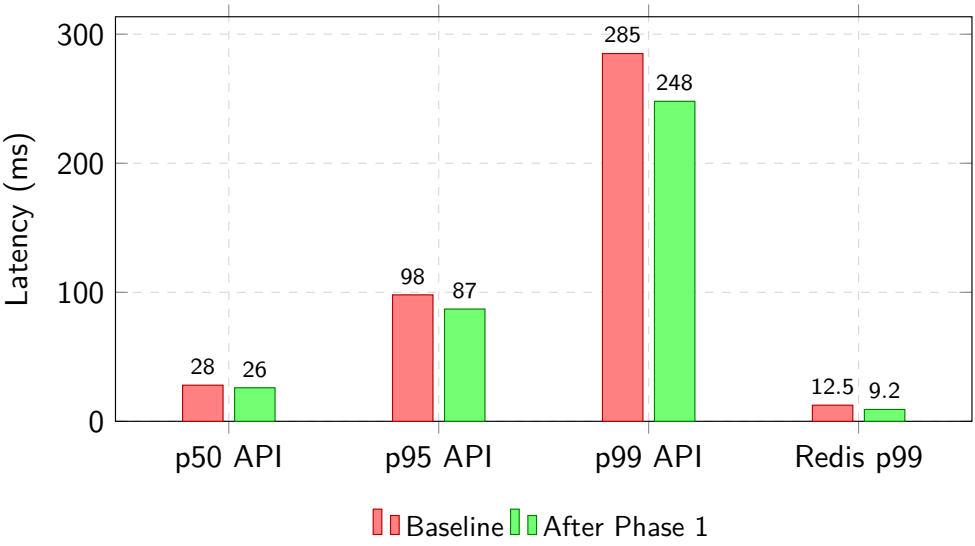
Decision Matrix

Data class	Cache?	Primary	Coalescing	Notes
Merchant config (read)	Tier-2	—	Single-flight	Cache-eligible
Routing table (read)	Tier-2	—	Single-flight	Cache-eligible
Feature flag (read)	Tier-2	—	Single-flight	Cache-eligible
Inventory (display)	Tier-2	—	Single-flight	Approximate
Inventory (decrement)	No	Lua atomic	Single-flight	Hard limit
Account balance debit	No	Distributed lock	Single-flight	Fence token
Reservation slot	No	Lua atomic	Single-flight	Per-slot key
Soft hold (cart)	Tier-2	PN-Counter	—	Over-alloc OK
Merchant config (write)	Inval.	OCC versioned	—	Low contention

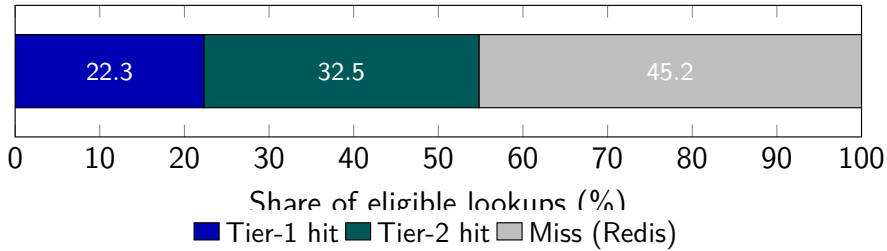
Comparison of the Five Primitives

Property	Lock	Lua	OCC	CRDT	Single-flight
Hard upper bound	Yes	Yes	Yes	No	N/A
Per-key throughput	Low	Very high	Medium–high	Very high	N/A
Latency overhead	1–3 ms	$<$1 ms	$<$1 ms	0 ms	0–1 ms
Behaviour under partition	Blocks/fails	Fails fast	Fails fast	Continues	Continues
Expressiveness	High	Lua only	SQL/CAS	Counter	N/A
Composes with cache	Bypass	Bypass + invalidate	Versioned cache	Cached	Front of cache

Tail-Latency Reductions (Phase 1)



Cache Hit-Rate Breakdown



- Combined hit rate **54.8%** on eligible lookups
- ∼2,800 cross-node calls/sec eliminated cluster-wide

Phase 2: Contention Layer in Production (30 days)

Workload	Primitive	Contention rate	p95 added
Inventory decrement (flash sale)	Lua + single-flight	1,400 contended/s	0.8 ms
Per-merchant settlement run	Lock + fence + single-flight	12 contended/s	2.6 ms
Reservation slot booking	Lua + single-flight	320 contended/s	1.1 ms
Cart soft hold	PN-Counter (no coord.)	N/A	0.0 ms

Correctness

- **Zero** double-allocation incidents
- **Zero** cache-related financial discrepancies
- Soft-hold over-allocation: 4.7 units/day on a 100k base ($<$1% tolerance)

Conclusion

What we built

An integrated two-layer system: a **caching layer** for read-side latency reduction and a **contention layer** that closes the single-resource race the minor evaluation surfaced.

- Cache-safety framework formalises which data is safe to cache
- Five contention primitives, each with a clear use case
- Decision matrix selects the right primitive per data class
- No new infrastructure beyond existing Redis cluster

Headline numbers

–11.2% p95, **–13.0% p99**, **–26.2% Redis p99**,
4.1% cross-node calls eliminated, **zero** correctness incidents.

Future Work

- **Adaptive TTLs** that shrink under detected mutation pressure
- **Learned contention prediction** — pre-warm the right primitive before flash sales
- **Verified fencing** — TLA$^+$ / Coq mechanisation of the safety argument
- **Cross-DC extension** — geo-replicated Redis vs. leader-elected coordinators
- **Graceful degradation modes** when Redis is partially unavailable

Selected References (2024+)

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Thank You

Questions?

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